Albireo Vector (ATH-771)

Quantitative Multi-Strategy Alternative Investment Fund

Track Record: Mar 2023 – Jan 2026 (35 Months / 2.9 Years) | **Benchmark:** S&P 500 Total Return

ATHÉNÉE INVESTMENT

CAPITAL INTRODUCTION & THIRD-PARTY MARKETING

CAGR (Net)	Ann. Volatility	Sharpe Ratio	Max Drawdown	Alpha (Ann.)	Beta & Corr.
83.75%	**11.06%**	**5.69**	**-0.43%**	**68.98%**	**-0.28 | -0.29**
S&P 500: 22.81%	S&P 500: 11.56%	SPX: 1.85 | Sortino: 250.05	SPX: -8.25% | Calmar: 194.76	t-stat: 9.66	Beta t-stat: -1.77

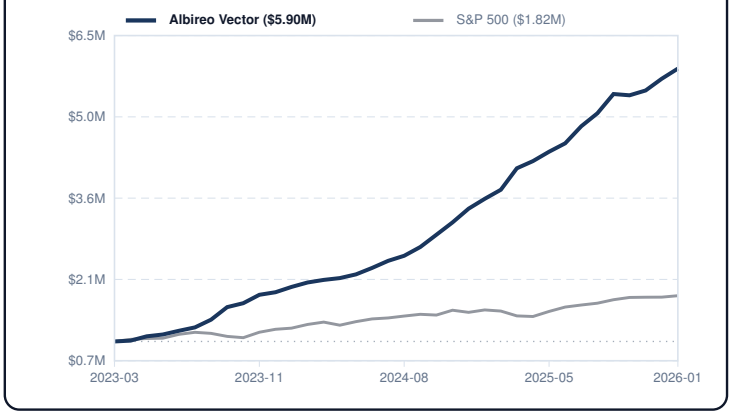
Performance, Risk & Hit Rate Indicators

Indicator	Fund	S&P 500
CAGR (Annualized)	**83.75%**	22.81%
Annualized Mean Return	62.96%	21.37%
Annualized Volatility	11.06%	11.56%
Proxy Sharpe Ratio ($r_f = 0\%$)	**5.69**	1.85
Annualized Sortino Ratio	250.05	3.92
Maximum Drawdown (MDD)	**-0.43%**	-8.25%
Calmar Ratio (CAGR / |MDD|)	194.76	2.76
Cumulative Total Return	489.71%	82.09%
Best / Worst Monthly	+16.18% / -0.43%	+9.13% / -5.63%
Positive Months (Hit Rate)	97.1% (34/35)	74.3% (26/35)
Crisis Hit Rate (Fund > 0 | SPX < 0)	**100.0% (9/9)**	–
Up-Mkt Loss (Fund < 0 | SPX > 0)	3.8% (1/26)	–

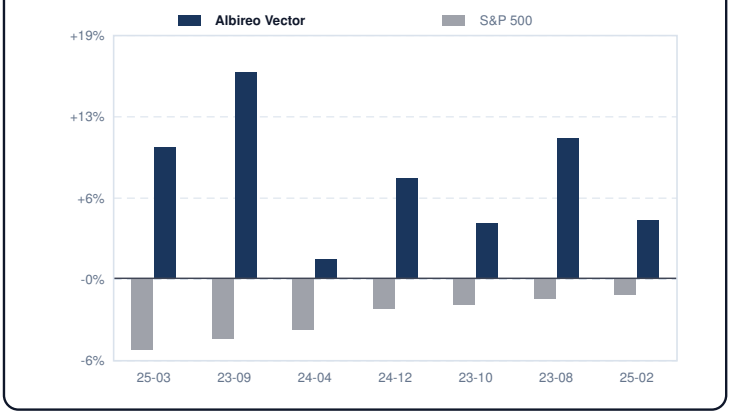
Alpha, Beta, Correlation & Tail Analysis (vs. S&P 500)

Quantitative Metric	Estimate / Diagnostic
Annualized Alpha (α) & t-stat	**68.98% (t=9.66)**
Market Beta (β_{SPX}) & t-stat	-0.28 (t=-1.77) ($R^2 = 8.7\%$)
Pearson Correlation (r) to S&P 500	**-0.29** (95% Blk CI: [-0.59, 0.20])
Spearman Rank Correlation (ρ)	-0.16 (95% Blk CI: [-0.46, 0.19])
Upside / Downside Capture Ratio	Up: **1.36** | Down: **-2.86**
Cond. Up-Market (SPX > 0)	$\alpha = 41.0\%,\ \beta = 0.32$
Cond. Down-Market (SPX < 0)	$\alpha = 56.1\%,\ \beta = -1.07$
Tail Corr. (SPX $< Q_1$, Worst 25%)	$r = -0.42,\ \rho = -0.33\ (n = 9)$
Tail Corr. (SPX $< D_1$, Worst 10%)	$r = -0.36,\ \rho = -0.60\ (n = 4)$
Worst 20% SPX Months Avg. Return	Fund: +7.98% vs SPX: -3.12%
Cross-Asset Beta (WTI / BTC)	WTI β: 0.12 | BTC β: 0.01
Ljung-Box Serial Correlation	$p_1 = 0.22,\ p_3 = 0.30$ (Pass)

Cumulative Growth of $1M USD (Fund vs. S&P 500)



Worst 20% S&P 500 Months



Historical Monthly & Yearly Net Return Matrix (%)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Yearly
2023	-	-	1.4%	7.6%	3.0%	5.8%	5.2%	11.1%	16.2%	4.4%	9.0%	2.5%	**88.3%**
2024	5.0%	4.0%	2.3%	1.5%	3.1%	5.2%	5.5%	3.7%	6.2%	8.2%	7.6%	7.8%	**79.6%**
2025	5.2%	4.6%	10.3%	3.1%	4.0%	3.4%	6.7%	4.8%	6.8%	-0.4%	1.6%	3.8%	**68.9%**
2026	3.2%	-	-	-	-	-	-	-	-	-	-	-	**3.2%**

Fund Terms & Commercial Overview

Vehicle:	Commingled Fund	Return Basis:	Net of Fees	Fee Structure:	2% Mgmt / 20% Perf	Liquidity:	Quarter Liquidity
Lock-Up Period:	No Lockup	Min. Investment:	1M USD	Current AUM:	250M	Domicile:	Cayman Islands

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Capital Introduction & Third-Party Marketing Memorandum. Strictly for Institutional & Qualified Investors. Please see Page 2 for Risk & Legal Disclosures.

Risk Disclosures, Legal Notice & Quantitative Appendix

Fund Reference: Albireo Vector (ATH-771) (Mar 2023 - Jan 2026 (35 Months / 2.9 Years))

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Appendix A: Extended Statistical, Bootstrap & Factor Diagnostics

Diagnostic Family	Methodology / Statistical Specification	Empirical Value
Pearson Correlation CI	95% Confidence Interval (IID Bootstrap vs. Moving Block $B = 3$, 1,000 iters)	Std: [-0.66, 0.26] | Blk: [-0.59, 0.20]
Spearman Rank Corr. CI	95% Confidence Interval (Rank Monotonic Co-Movement, IID vs. Block $B = 3$)	Std: [-0.53, 0.27] | Blk: [-0.46, 0.19]
Worst-Case Co-Dependence	97.5th Percentile Upper-Bound Block Bootstrap Correlation (PSD Projection)	$\rho_{\text{wc}} = 0.20$
Serial Correlation (Lag 1 & 3)	Ljung-Box Q-Statistic p-values testing for return smoothing / illiquidity bias	$p_1 = 0.22,\ p_3 = 0.30$ (Pass)
Conditional Up-Market OLS	Annualized Alpha & Beta estimated during positive S&P 500 months ($n = 26$)	$\alpha = 41.0\%,\ \beta = 0.32\ (t_\alpha = 4.55)$
Conditional Down-Market OLS	Annualized Alpha & Beta estimated during negative S&P 500 months ($n = 9$)	$\alpha = 56.1\%,\ \beta = -1.07\ (t_\alpha = 1.70)$
Commodity Factor (WTI)	OLS Regression against Front-Month WTI Crude Oil Monthly Returns	$\beta = 0.12\ (t = 1.56),\ \alpha = 63.3\%$
Digital Asset Factor (BTC)	OLS Regression against Bitcoin (BTC-USD) Monthly Returns	$\beta = 0.01\ (t = 0.16),\ \alpha = 62.6\%$

Appendix B: Quantitative Framework & Indicator Definitions

- **CAGR & Annualized Volatility:** Compound Annual Growth Rate is computed geometrically as CAGR $= \left(\prod_{t=1}^{T}(1 + r_t)\right)^{12/T} - 1$. Annualized Volatility is sample standard deviation scaled by $\sqrt{12}$: $\sigma_{\text{ann}} = \hat{\sigma}_m\sqrt{12}$. Proxy Sharpe Ratio assumes $r_f = 0\%$ ($\bar{r}_{\text{ann}}/\sigma_{\text{ann}}$).
- **Maximum Drawdown (MDD) & Calmar Ratio:** Maximum peak-to-trough decline of cumulative compounded wealth $W_t = \prod_{s\le t}(1 + r_s)$, defined as MDD $= \min_t\left((W_t - \max_{s\le t} W_s)/\max_{s\le t} W_s\right)$. Calmar Ratio is CAGR/|MDD|.
- **OLS Alpha (α), Beta (β) & Capture Ratios:** Estimated via Ordinary Least Squares $r_{\text{fund},t} = \alpha_m + \beta r_{\text{SPX},t} + \varepsilon_t$, where $\alpha_{\text{ann}} = 12 \times \alpha_m$. Upside (Downside) Capture measures the ratio of the fund's mean return to the S&P 500's mean return during positive (negative) S&P 500 months.
- **Tail Correlations & Moving Block Bootstrap:** Tail correlations evaluate linear (r) and rank (ρ) co-movement during S&P 500 drawdowns below the 25th percentile (Q_1) and 10th percentile (D_1). Moving Block Bootstrap ($B = 3$ months) preserves local serial dependencies when constructing 95% confidence intervals.

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